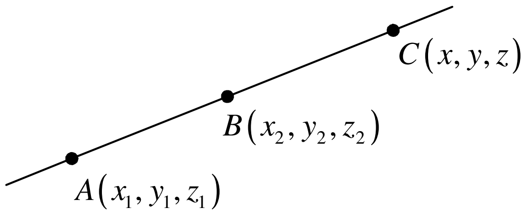
§12.5 Equations of Lines and Planes

Homework： 1,5,11,15,19,37,41,49,59,63,67

(1) Equation of lines： Given A and B, find equation of the line AB.



$\overrightarrow{AB} = <x_2 - x_1, y_2 - y_1, z_2 - z_1>$, $\overrightarrow{AC} = <x - x_1, y - y_1, z - z_1>$

$=: <a, b, c>$

$\overrightarrow{AB} // \overrightarrow{AC} \Rightarrow \exists\, t \in R$ s.t. $\overrightarrow{AC} = t\overrightarrow{AB}$.

$\Rightarrow$

(i) Parametric equation

$$\begin{cases} x = x_1 + t(x_2 - x_1) \\ y = y_1 + t(y_2 - y_1) \\ z = z_1 + t(z_2 - z_1) \end{cases} , \quad t \in R .$$

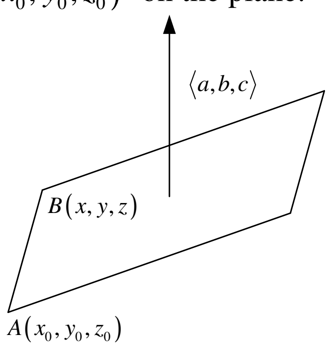
(ii) Symmetric equation

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} , \quad abc \neq 0$$

(If $a = 0$, $bc \neq 0 \Rightarrow x = x_1$, $\dfrac{y - y_1}{b} = \dfrac{z - z_1}{c}$)

Remark： $<a, b, c>$ is called the direction of the line AB.

(2) Equation of the Plane： Given the normal direction of the plane $<a, b, c>$ and a point (x_0, y_0, z_0) on the plane.

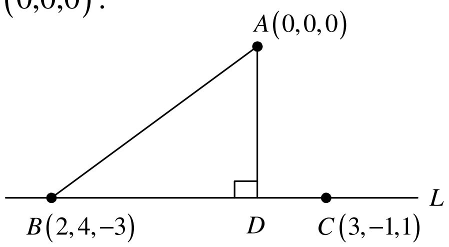


$\Rightarrow$ the equation of the plane：

(i) $\langle a,b,c\rangle \langle x-x_0, y-y_0, z-z_0\rangle = 0$

(ii) $ax+by+cz = ax_0+by_0+cz_0 \left(=d\right)$

Example 1： Line L through $(2,4,-3)$, $(3,-1,1)$. Find the distance between L and $(0,0,0)$.



Solution：

$\overrightarrow{BA} = \langle -2,-4,3\rangle$ and $\overrightarrow{BC} = \langle 1,-5,-4\rangle$

Vector projection of $\overrightarrow{BA}$ on $\overrightarrow{BC}$.

$$= \overrightarrow{BD} = \left(\frac{\overrightarrow{BA}\cdot\overrightarrow{BC}}{\left|\overrightarrow{BC}\right|}\right)\frac{\overrightarrow{BC}}{\left|\overrightarrow{BC}\right|} = \left\langle \frac{1}{7}, \frac{-5}{7}, \frac{-4}{7}\right\rangle$$

$$\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{BD} = \langle 2,4,-3\rangle + \left\langle \frac{1}{7}, \frac{-5}{7}, \frac{-4}{7}\right\rangle$$

$$= \left\langle \frac{15}{7}, \frac{23}{7}, \frac{-25}{7}\right\rangle$$

The distance $= \left|\overrightarrow{AD}\right|$.

Example 2： Find the equation of the plane through $(1,3,2),(3,-1,6)$ and $(5,2,0)$.

Solution：

$$\text{法向量} = <2,-4,4>\times<4,-1,-2>$$
$$= <12,20,14>$$
$$= 2<6,10,7>$$
$$<6,10,7>\cdot<x-5\ ,\ y-2\ ,\ z-0> = 0$$
$$\Rightarrow 6x+10y+7z = 50.$$

Example 3：

(i) Find the angles between two planes $P_1 : x+y+z=1$ and $P_2 : x-2y+3z=1$.

(ii) Let $l = P_1 \cap P_2$. Find the equation of l.

Solution：

(i) 平面夾角=法向量夾角

$$\Rightarrow \cos\theta = \frac{<1,1,1>\cdot<1,-2,3>}{\sqrt{3}\ \sqrt{14}} = \frac{2}{\sqrt{42}} = \frac{\sqrt{42}}{21}$$

$$\Rightarrow \theta = \cos^{-1}\frac{\sqrt{42}}{21}$$

(ii) l 的方向和 P_1 ， P_2 平面的法向量垂直

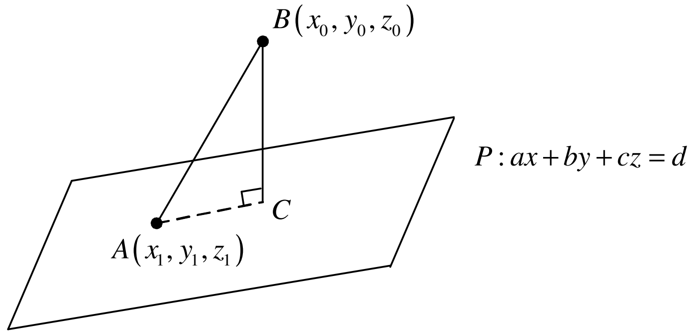
$\Rightarrow <1,1,1>\times<1,-2,3> = <5,-2,-3> =$ 直線 L 的方向.

$$\begin{cases} x+y+z=1 \\ x-2y+3z=0 \end{cases} \Rightarrow \text{令 } z=0 \ \Rightarrow \ y=\frac{1}{3} \text{ and } x=\frac{2}{3}.$$

$$\Rightarrow \frac{x-\frac{2}{3}}{5} = \frac{y-\frac{1}{3}}{-2} = \frac{z}{-3}.$$

Example 4： Find the distance between two parallel planes $10x+2y-2z=5$ and $5x+y-z=1$.

Solution：



$$\overrightarrow{BC} = \left| \text{ Scalar projection of } \overrightarrow{BA} \text{ on } \overrightarrow{BC} \right|.$$

$$= \left| \frac{\overrightarrow{BA}\cdot\overrightarrow{BC}}{\overrightarrow{BC}} \right|$$

$$= \left| \frac{ax_0+by_0+cz_0-d}{\sqrt{a^2+b^2+c^2}} \right|$$

$=$點到面之距離公式 .

$$\left(\overrightarrow{BA} = \langle x_1-x_0 \ , \ y_1-y_0 \ , \ z_1-z_0 \rangle \ , \ \overrightarrow{BC} = \langle a,b,c \rangle \right)$$

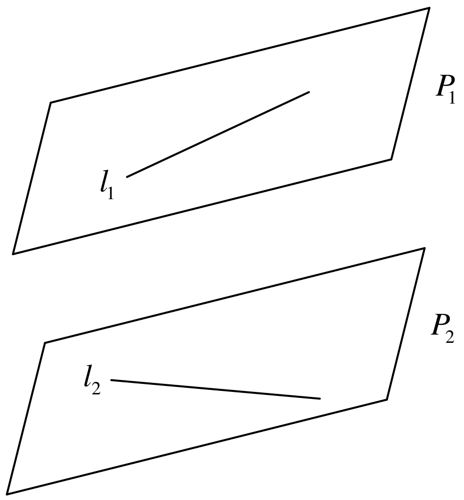
Now, 令 $x=0$, $y=\dfrac{5}{2}$, $z=0$,

則＝平行平面之距離＝ $\dfrac{\frac{5}{2}}{\sqrt{27}}=\dfrac{5\sqrt{3}}{18}$.

Example 5： Find the distance between two skew lines

$$l_1:\begin{cases}x=1+t\\ y=-2+3t\\ z=4-t\end{cases},\ t\in R\ ,\quad l_2:\begin{cases}x=2s\\ y=3+s\\ z=-3+4s\end{cases},\ s\in R\ .$$

Solution：



找 $P_1 // P_2$ 且 $l_1\in P_1$, $l_2\in P_2$.

$\Rightarrow$ P_1 and P_2 的法向量 $n\perp l_1$, l_2 .

$\Rightarrow n=\langle 1,3,1\rangle\times\langle 1,1,4\rangle=\langle 11,-3,-2\rangle$

將 P_1 , P_2 方程式找出

$\Rightarrow d(l_1,l_2)=d(P_1,P_2)$.

§12.6 Cylinders and Quadrate Surfaces

Homework： 9,21-28,29,31,35,43,46,48

(1) Cylinders： A surface consisting of all lines (called rulings) that are // to a given line and pass through a given plane curve.

Example 1： (i) $z = x^2$ (Parabolic cylinder)
(ii) $x^2 + y^2 = 1$

Remark： If one of the variables x , y or z is missing from the equation of a surface , then the surface is a cylinder.

(2) Quadratic Surfaces：

$$ax^2 + by^2 + cz^2 + dxy + eyz + fxz + gx + hy + iz + j = 0$$

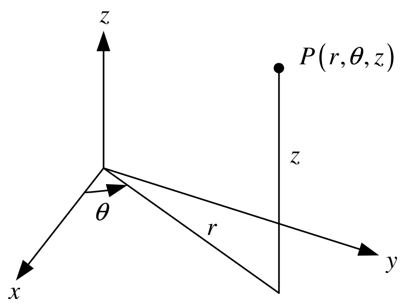
See P.836 of the test book for the classification of quadratic surface.

(3) Cross-sections or traces

§12.7 Cylindrical and Spherical Coordinates

Homework： 21,25,29,35,39

(i) Cylindrical Coordinate



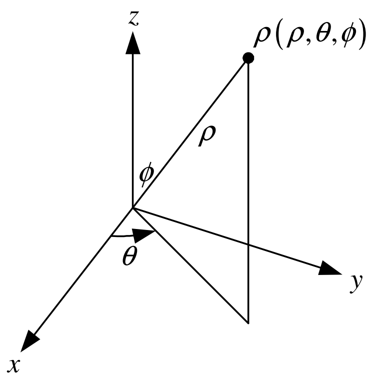
$(r,\theta,z) \leftrightarrow (x,y,z)$

$x = r\cos\theta$

$y = r\sin\theta$

$z = z$

(ii) Spherical Coordinates



$(x,y,z) \leftrightarrow (\rho,\theta,\phi)$

$x = \rho\sin\phi\cos\theta$

$y = \rho\sin\phi\sin\theta$

$z = \rho\cos\phi$